VICTORIA UNIVERSITY OF WELLINGTON
TE HERENGA WAKA

Last Name .....

First Name .....

Student ID Number .....

Course Title **TECHNIQUES IN MACHINE LEARNING**

Course Code and No. **AIML 231**

| Question Number | Marks Available | Examiners' use only |
|---|---|---|
| Q1 | 8 | |
| Q2 | 22 | |
| Q3 | 20 | |
| Q4 | 13 | |
| Q5 | 25 | |
| Q6 | 7 | |
| Q7 | 5 | |
| **TOTAL** | **100** | |

**This booklet must be handed in at the end of the examination.**

## PLEASE READ THESE INSTRUCTIONS CAREFULLY

1. Fill in clearly all blanks for insertion of name, student number, course title and code.
2. Be sure to read the instructions at the head of each section very carefully; also any further instructions for any specific question.
3. Cross out neatly any work not intended for the examiner.
4. Supplementary books are available from the supervisor.
5. Tie this and any supplementary books together and hand the whole script to the supervisor.
6. Any script or portion of a script which is removed from the examination room, whether inadvertently or otherwise, will not be marked.
7. Rough work must be done in the answer book and any work not intended for the examiner should be crossed out.
8. Do not communicate with the examiner, either on the script or in any manner except through the Examinations Office, Victoria University of Wellington.

VICTORIA UNIVERSITY OF WELLINGTON
TE HERENGA WAKA

**EXAMINATIONS – 2024**

**TRIMESTER 1**

**FRONT PAGE**

**AIML 231/DATA 302**

**Techniques in Machine Learning**
**12/06/2024**

**Time Allowed:** TWO HOURS

**CLOSED BOOK**

**Permitted materials:** Only silent non-programmable calculators are permitted in this test. Non-electronic foreign language translation dictionaries may be used.

**Instructions:** The exam will be marked out of a total of 100 marks.
Attempt all the questions.

# Questions


1. Machine Learning Basics [ 8]
2. Classification [22]
3. Machine Learning Pipeline and Data Preparation [20]
4. Regression and Clustering [13]
5. Neural Networks [25]
6. Search and Evolutionary Computation [ 7]
7. Advanced Topics [ 5]

1. Machine learning Basics **(8 marks)**

(a) **(3 marks)** Briefly explain the concept of Unsupervised Machine Learning. Provide two examples of Unsupervised Machine Learning tasks.

(b) **(2 marks)** Briefly explain “Curse of Dimensionality” in machine learning.

(c) **(3 marks)** Identify three key differences between Supervised Machine Learning and Reinforcement Learning.

2. Classification **(22 marks)**

(a) **(2 marks)** Classification and regression are two important supervised machine learning tasks. Briefly state a key difference between classification and regression.

(b) **(3 marks)** In supervised machine learning systems such as a classification system, a dataset is typically separated into a training set, a test set, and a validation set. Briefly describe the role of each of them.

(c) **(5 marks)** Briefly describe the key idea of K-fold cross validation method used in machine learning experiments. Briefly explain when cross validation methods should be used.

(d) **(3 marks)** Considering the use of the K Nearest Neighbour (KNN) classifier, when the hyper-parameter $K$ is set to a large value, *list two problems* that the classifier will experience.

(e) **(4 marks)** In a medical diagnostic test for a disease, out of 200 patients, 40 are known to have the disease, and 160 do not. The test correctly identifies 28 patients with the disease but also incorrectly identifies 20 patients as having the disease when they do not. Calculate the **false negative rate** and **true negative rate** of this medical diagnostic system. Show your working.

(f) **(5 marks)** Consider the following dataset describing 10 menu items from a restaurant, of which 5 are popular with customers, and 5 are not. They are described by three attributes: whether they are spicy or mild, whether they have sauce or not, and whether the protein base is vegetarian, beef, or chicken.

Given the entropy equations as follows, calculate the information gain of using the spice feature to split this dataset. Show your working.

$$H(Y) = -\sum_{i=1}^{k} P(Y = y_i) \log_2 P(Y = y_i)$$
$$H(Y|X) = -\sum_{j=1}^{v} P(X = x_j) \sum_{i=1}^{k} P(Y = y_i|X = x_j) \log_2 P(Y = y_i|X = x_j)$$

| Spice | Sauce | Protein | Category |
|---|---|---|---|
| spicy | yes | beef | Popular |
| spicy | yes | beef | Popular |
| spicy | yes | vegetarian | Popular |
| mild | no | beef | Popular |
| spicy | no | beef | Popular |
| spicy | yes | chicken | unpopular |
| spicy | yes | vegetarian | unpopular |
| spicy | no | beef | unpopular |
| mild | no | beef | unpopular |
| mild | no | chicken | unpopular |

3. Machine Learning Pipeline and Data Preparation **(20 marks)**

(a) **(3 marks)** Cross Industry Standard Process for Data Mining (CRISP-DM) is a widely used machine learning process. CRISP-DM has six main phases in its model. List the six phases of CRISP-DM.

(b) **(2 marks)** Discretisation is the process of converting from a continuous feature to a discrete feature. Briefly describe the *Uniform* discretisation method and the *Quantile* discretisation method.

(c) **(3 marks)** Nick needs to perform classification on a give dataset. Unfortunately, the features have different scales. Thus, he normalises the dataset first to ensure all the features are in the range [0, 1]. After that, he splits the dataset into a training set and a test set. Finally, he trains an SVM classification algorithm on the training set. Is Nick's approach an appropriate way to do classification? Justify your answer.

(d) **(3 marks)** Feature construction and feature selection are two main approaches to improve the quality of features. What is the main difference between feature construction and feature selection?

(e) **(4 marks)** Sequential Forward Feature Selection (SFFS) is a commonly used feature selection algorithm. In this question, you will apply SFFS to select *3* out of the *4* original features: $\{f_1, f_2, f_3, f_4\}$. You will use classification accuracies to determine which feature to add at each step of SFFS. The classification accuracies of all the possible feature subsets are given in the following table. Justify which feature to add at *each step* of SFFS and when to *terminate* the algorithm. Show your working.

| | | | | | |
|---|---|---|---|---|---|
| **Subset** | $\{f_1\}$ | $\{f_2\}$ | $\{f_3\}$ | $\{f_4\}$ | $\{f_1, f_2\}$ |
| **Performance** | 0.6 | 0.2 | 0.7 | 0.65 | 0.75 |
| **Subset** | $\{f_1, f_3\}$ | $\{f_1, f_4\}$ | $\{f_2, f_3\}$ | $\{f_2, f_4\}$ | $\{f_3, f_4\}$ |
| **Performance** | 0.55 | 0.8 | 0.6 | 0.2 | 0.76 |
| **Subset** | $\{f_1, f_2, f_3\}$ | $\{f_1, f_2, f_4\}$ | $\{f_1, f_3, f_4\}$ | $\{f_2, f_3, f_4\}$ | $\{f_1, f_2, f_3, f_4\}$ |
| **Performance** | 0.62 | 0.9 | 0.8 | 0.5 | 0.7 |

(f) **(5 marks)** Principal Component Analysis (PCA) is a well-known feature construction method where the newly built features are called principal components.

- (1) Feature construction can be divided into 3 main categories: wrapper, filter, and embedded approaches. Which category does PCA belong to? Justify your answer.
- (2) Suppose a dataset has *10* original features, what is the maximum number of principal components that we can build?
- (3) List two main limitations of PCA.

4. Regression and Clustering **(13 marks)**

(a) **(2 marks)** The overall goal of Linear Regression is to build a hyperplane that can fit a given training set. The hyperplane is represented by a weight vector which needs to be optimised. Suppose the training set has *10* features, how many weights are there in the weight vector? Justify your answer.

(b) **(3 marks)** Ridge Regression and Lasso Regression extend Linear Regression by having an additional regularisation component. Among Ridge Regression and Lasso Regression, which one can be used as an *embedded feature selection* approach? Briefly justify your answer.

(c) **(8 marks)** K-means and Agglomerative Clustering are two well-known clustering algorithms which we discussed in our class.

- (1) Briefly describe the main steps of K-means.
- (2) List two limitations of K-means.
- (3) Suppose there is a large dataset containing 1,000,000 instances. Among the two above clustering methods, which one is more efficient to perform clustering on the given large dataset? Justify your answer.

5. Neural Networks **(25 marks)**

(a) **(1 mark)** What role do neurons in a neural network play?

- A) They serve as database storage units.
- B) They process inputs and produce outputs based on those inputs.
- C) They reduce data dimensionality automatically.
- D) They act as filters for noise in data.

(b) **(1 mark)** What is the primary purpose of an activation function in neural networks? (1 mark)

- A) To normalise the input data
- B) To allow the model to learn non-linear patterns
- C) To reduce the dimensionality of the input data
- D) To increase the speed of convergence

(c) **(2 marks)** Match the following terms to their appropriate descriptions: (2 marks)

- A) Batch Gradient Descent
- B) Stochastic Gradient Descent
- C) Mini-batch Gradient Descent

1. A single update of the model's weights during training
2. Update model weights using all data iteratively
3. Using small subsets of the dataset to update model weights iteratively

Example answer: A-1, B-2, ...

(d) **(1 mark)** List one limitation of a single-layer perceptron that are overcome by using multiple layers.

(e) **(5 marks)** Calculate the total number of weights in the Multi-Layer Perceptron (MLP).

Consider a Multi-Layer Perceptron (MLP) with the following architecture:

- An input layer with 10 neurons
- Two hidden layers:
  - The first hidden layer has 20 neurons
  - The second hidden layer has 15 neurons
- An output layer with 5 neurons

Assume that each neuron in a layer is connected to every neuron in the previous layer, and each connection represents a weight. Also, assume that each neuron in the hidden and output layers has a bias term, which is considered a (single) weight. Find the total number of weights in the MLP; Show your working.

(f) **(3 marks)** Consider the ReLU (Rectified Linear Unit) and sigmoid activation functions, which activation function is more commonly used in the output layer for binary classification tasks? Briefly explain your choice.

(g) **(3 marks)** Briefly discuss the role of the loss function in neural networks and list two common loss functions.

(h) **(2 marks)** Briefly explain why gradient descent is called a 'descent' method.

(i) **(2 marks)** What is the aim of backpropagation in Neural Network Learning?

(j) **(2 marks)** List two strategies that can be used to prevent or mitigate overfitting in a Neural Network.

(k) **(3 marks)** Briefly explain how automatic differentiation facilitates the training process of neural network models?

6. Search and Evolutionary Computation **(7 marks)**

(a) **(2 marks)** Briefly describe a limitation of hill climbing. List a search method that helps to overcome this limitation.

(b) **(2 marks)** Briefly explain the role of the heuristic in A* search.

(c) **(3 marks)** Match the following terms in Evolutionary Computation to their correct descriptions.

- A Selection
- B Elitism
- C Mutation
- D Fitness
- E Population
- F Crossover

1. A measure of how well an individual in the population performs with respect to the given problem.
2. A genetic operator operated by making small random changes to the genetic material of the individuals.
3. A collection of individual solutions that are evaluated and evolved through the application of genetic operators.
4. The process of choosing which individuals get to breeding based on their fitness.
5. A genetic algorithm technique where a few of the best individuals are guaranteed to survive to the next generation.
6. A genetic operator used to combine the genetic information of two parent individuals to generate new offspring.

Example answer: A-1, B-2, ...

7. Advanced Topics **(5 marks)**

(a) **(2 marks)** Markov Decision Process (MDP) is widely used as the mathematical model of reinforcement learning problems. Briefly describe what the state transition function is in an MDP.

(b) **(3 marks)** Identify two optimization methods that can effectively address the hyper-parameter optimization problem with both discrete and continuous hyper-parameters in an Automated Machine Learning (AutoML) system.

* * * * * * * * * * * * * *